

Table 1. **The robot-evaluation benchmark landscape: 160 web-verified benchmarks, compared on four axes.** Grouped into four evaluation lanes and sub-grouped by capability. Columns: **Year**; **Capability** (what the benchmark stresses); **Eval** (evaluation mode: open-loop prediction/generation quality, closed-loop task success, or a prediction-to-action bridge); and **VLA vs WM** (does the benchmark build a native direct-policy vs world-model contrast: ✓ explicit, ~ partial, – model-agnostic). Every entry is a real, individually citeable benchmark.

Benchmark	Year	Capability	Eval	VLA vs WM
Policy / manipulation suites (37)				
<i>Manipulation</i>				
ARNOLD [?]	2023	language-grounded continuous-state 3D manipulation	closed	–
BiGym [?]	2024	mobile bimanual manipulation	closed	–
FMB [?]	2024	real-world functional multi-step assembly manipulation	closed	–
Franka Kitchen [?]	2019	multi-task kitchen manipulation	closed	–
FurnitureBench [?]	2023	real-world furniture assembly	closed	–
GRUtopia [?]	2024	city-scale embodied navigation and manipulation	closed	–
KitchenShift	2021	kitchen distribution-shift generalization	closed	–
ManiSkill2 [?]	2023	generalizable skills	closed	–
PerAct [?]	2022	language-conditioned 6-DoF single-arm manipulation	closed	–
PerAct2 [?]	2024	bimanual language-conditioned manipulation	closed	–
Ravens [?]	2020	vision-based pick-and-place rearrangement	closed	–
RGB-Stacking [?]	2021	vision-based stacking of diverse shapes	closed	–
RLBench [?]	2020	manipulation breadth	closed	–
RoboAgent [?]	2023	multi-task manipulation with RoboSet	closed	–
robomimic [?]	2021	single-arm manipulation from human demonstrations	closed	–
robosuite [?]	2020	modular simulated single-arm manipulation tasks	closed	–
RoboTwin [?]	2024	dual-arm bimanual manipulation	closed	–
VLMbench [?]	2022	vision-and-language compositional manipulation	closed	–
<i>Generalization & robustness</i>				
GemBench [?]	2024	generalization levels	closed	–
LIBERO [?]	2023	short-horizon + transfer	closed	–
RoboArena [?]	2025	real-world generalization	closed	–
SIMPLER / SimplerEnv [?]	2024	generalization (real vs sim)	closed	–
THE COLOSSEUM [?]	2024	generalization / robustness	closed	–
VIMA-Bench [?]	2023	multimodal-prompt generalization	closed	–
<i>Other</i>				
CortexBench [?]	2023	pretrained visual representations for embodied control	closed	–
GenManip [?]	2025	LLM-scene tasks	closed	~
HandoverSim [?]	2022	human-to-robot object handover	closed	–
Meta-World [?]	2020	multi-task / meta-RL	closed	–
RoboHive [?]	2023	unified robot-learning environments	closed	–
<i>Dexterous manipulation</i>				
Bi-DexHands [?]	2022	bimanual dexterous manipulation	closed	–
DeformableGym	2023	3D deformable-object grasping	closed	–
DexArt [?]	2023	dexterous articulated-object manipulation	closed	–
UniDexGrasp [?]	2023	universal dexterous grasping	closed	–
<i>Long-horizon & planning</i>				
CALVIN [?]	2021	long-horizon language	closed	–
RoboCasa [?]	2024	long-horizon (data scaling)	closed	–
VLABench [?]	2025	long-horizon reasoning	closed	–
<i>Deformable-object manipulation</i>				
SoftGym [?]	2020	deformable-object manipulation	closed	–
Embodied navigation, planning & social (85)				
<i>Long-horizon & planning</i>				

Table 1 (continued)

Benchmark	Year	Capability	Eval	VLA vs WM
Alexa Arena [?]	2023	task-planning	closed	—
ALFRED [?]	2019	long-horizon, partial-obs	closed	—
BEHAVIOR-1K [?]	2024	long-horizon household	closed	—
CoELA (C-WAH/TDW-MAT) [?]	2023	long-horizon	closed	—
DialFRED [?]	2022	task-planning	closed	—
EgoPlan-Bench [?]	2023	task-planning	closed	—
EmbodiedBench [?]	2025	long-horizon	closed	—
EmbodiedCity [?]	2024	long-horizon	closed	—
EmbodiedEval [?]	2025	long-horizon	closed	—
HomeRobot OVMM [?]	2023	long-horizon	closed	—
LEGENT [?]	2024	long-horizon	closed	—
LH-VLN [?]	2024	long-horizon	closed	—
LoHoRavens [?]	2023	long-horizon	closed	—
LoTa-Bench [?]	2024	task-planning	closed	—
MFE-ETP [?]	2024	task-planning	closed	—
PARTNR [?]	2024	task-planning	closed	—
PlanBench [?]	2022	task-planning	closed	—
SafeAgentBench [?]	2024	task-planning	closed	—
SMART-LLM [?]	2023	task-planning	closed	—
TDW-Transport [?]	2021	task-planning	closed	—
TidyBot [?]	2023	task-planning	closed	—
VirtualHome [?]	2018	task-planning	closed	—
Watch-And-Help [?]	2020	long-horizon	closed	—
<i>Social & multi-agent</i>				
CrowdNav [?]	2018	social	closed	—
Habitat 3.0 [?]	2023	social / multi-agent	closed	—
HuNavSim [?]	2023	social	closed	—
JRDB [?]	2019	social	closed	—
JRDB-Act [?]	2021	social	closed	—
Melting Pot [?]	2021	social	closed	—
Overcooked-AI [?]	2019	social	closed	—
RoboSense [?]	2024	social	closed	—
SEAN [?]	2020	social	closed	—
SMAC [?]	2019	social	closed	—
SocialGym 2.0 [?]	2023	social	closed	—
SocNavBench [?]	2021	social navigation	closed	—
<i>Navigation</i>				
Gibson Env [?]	2018	navigation	closed	—
GOAT-Bench [?]	2024	multi-modal lifelong navigation	closed	—
Habitat [?]	2019	navigation	closed	—
Habitat 2.0 [?]	2021	rearrangement	closed	—
HM3D [?]	2021	navigation	closed	—
iGibson [?]	2020	interactive navigation	closed	—
Matterport3D [?]	2017	navigation	closed	—
ObjectNav [?]	2020	object-goal navigation	closed	—
ProcTHOR [?]	2022	navigation	closed	—
RoboTHOR [?]	2020	navigation / sim-to-real	closed	—
SoundSpaces [?]	2020	audio-visual navigation	closed	—
<i>Autonomous driving</i>				
Bench2Drive [?]	2024	driving	closed	—
CARLA [?]	2017	driving	closed	—
DriveArena [?]	2024	driving	closed	✓
MetaDrive [?]	2021	driving	closed	—
NAVSIM [?]	2024	driving	closed	—
nuScenes [?]	2019	driving	closed	—
Waymax [?]	2023	driving	closed	—
Waymo Open [?]	2019	driving	closed	—
<i>Counterfactual reasoning</i>				
ACRE [?]	2021	counterfactual	open	✓
CausalCity [?]	2021	counterfactual	closed	—
CausalVQA [?]	2025	counterfactual	open	✓

Table 1 (continued)

Benchmark	Year	Capability	Eval	VLA vs WM
CausalWorld [?]	2020	counterfactual	closed	✓
CoPhy [?]	2019	counterfactual	open	✓
CRAFT [?]	2020	counterfactual	open	✓
Filtered-CoPhy [?]	2022	counterfactual	open	✓
<i>Locomotion</i>				
Barkour [?]	2023	legged locomotion	closed	–
Brax [?]	2021	locomotion	closed	–
DERL [?]	2021	locomotion	closed	–
HumanoidBench [?]	2024	humanoid locomotion	closed	–
Isaac Gym [?]	2021	locomotion	closed	–
Legged Gym [?]	2021	legged locomotion	closed	–
<i>Theory of mind</i>				
AGENT [?]	2021	theory-of-mind	open	✓
BIB [?]	2021	theory-of-mind	open	✓
PHASE [?]	2021	theory-of-mind	closed	–
SocialAI [?]	2021	theory-of-mind	closed	–
ToMi	2019	theory-of-mind	closed	–
<i>Vision-language navigation</i>				
R2R [?]	2018	VLN	closed	–
REVERIE [?]	2020	VLN	closed	–
RxR [?]	2020	VLN	closed	–
Touchdown [?]	2019	outdoor VLN	closed	–
VLN-CE [?]	2020	VLN	closed	–
<i>Hidden-state / object permanence</i>				
Bongard-HOI [?]	2022	hidden-state	open	–
ComPhy [?]	2022	hidden-state	open	✓
LIBERO-Mem [?]	2025	hidden-state / occlusion	closed	✓
TEACH [?]	2021	hidden-state (dialog)	closed	–
<i>Embodied question answering</i>				
OpenEQA	2024	EQA	closed	–
RoboVQA [?]	2023	EQA	closed	–
<i>Generalization & robustness</i>				
ALFWorld [?]	2020	abstract-vs-grounded transfer	closed	–
<i>Other</i>				
O-PIAAGETS	2022	object permanence	closed	–
World-model & video-generation evaluation (34)				
<i>Generation quality</i>				
CATER [?]	2019	compositional-action temporal reasoning	open	–
DEVIL [?]	2024	content-dynamics quality of T2V models	open	–
EVA-Bench [?]	2024	embodied video anticipation	open	–
EvalCrafter [?]	2023	generation quality (17-metric suite)	open	–
EWMBench [?]	2025	scene/motion/semantic quality	open	–
FETV [?]	2023	fine-grained text-to-video quality	open	–
IPV-Bench [?]	2025	impossible-video generation + understanding	open	–
StoryBench [?]	2023	continuous story visualization	open	–
T2V-CompBench [?]	2024	compositional text-to-video quality	open	–
T2VScore [?]	2024	text-to-video metric (alignment + quality)	open	–
TC-Bench [?]	2024	temporal compositionality in video generation	open	–
VBench [?]	2023	generation quality (16 disentangled dimensions)	open	–
VBench-2.0 [?]	2025	generation faithfulness	open	–
VideoCon [?]	2023	robust video-language alignment	open	–
VideoHalluciner [?]	2024	hallucination in video-language models	open	–
VideoScore [?]	2024	learned automatic quality metric	open	–
WorldModelBench [?]	2025	video-WM prediction quality	open	–
WorldScore [?]	2025	world-gen quality	open	–
<i>Physical reasoning</i>				
ContPhy [?]	2024	continuum (soft-body/fluid) physical reasoning	open	–
GRASP [?]	2023	grounding + intuitive physics in video MLLMs	open	–
IntPhys [?]	2018	intuitive physics (violation-of-expectation)	open	–
IntPhys 2 [?]	2025	intuitive physics in complex scenes	open	–
PHYBench [?]	2025	physical perception and reasoning	open	–

Table 1 (continued)

Benchmark	Year	Capability	Eval	VLA vs WM
PhyGenBench [?]	2024	physical-commonsense correctness in video gen	open	–
PhysBench [?]	2025	physical-world understanding for VLMs	open	–
Physics-IQ [?]	2025	physical realism	open	–
Physion [?]	2021	physical prediction from vision	open	–
Physion++ [?]	2023	physical prediction with online property inference	open	–
PhyWorldBench [?]	2025	physical realism in text-to-video	open	–
VideoPhy [?]	2024	physical commonsense in generated video	open	–
VideoPhy-2 [?]	2025	action-centric physical commonsense	open	–
<i>Counterfactual reasoning</i>				
CLEVRER [?]	2019	physical/causal video reasoning (counterfactual)	open	–
WorldPrediction [?]	2025	counterfactual action recognition	open	–
<i>Autonomous driving</i>				
Vista [?]	2024	driving world model	open	–
Bridges (prediction to action) (4)				
<i>Other</i>				
RoboWM-Bench [?]	2026	prediction executability	bridge	~
World-in-World [?]	2025	closed-loop task success of WMs	bridge	~
WorldArena [?]	2026	closed-loop WM arena	bridge	~
<i>Generation quality</i>				
WorldSimBench [?]	2024	video-to-action consistency	bridge	~